



CS & IT ENGINEERING



Algorithms

Dynamic Programming

Lecture No.- 03



By- Aditya sir

Recap of Previous Lecture



Topic

Topic

Approaches of DP

↳ 1) Top Down

2) Bottom Up

Topics to be Covered



Topic

Topic

Topic

DP $\rightarrow$ SSSP
Problems



About Aditya Jain sir

1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on LinkedIn where I share my insights and guide students and professionals.



Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW

TTTe

Sat/Sun → 19 + 20 July

8:30 AM



Topic : Dynamic Programming: (DP)

Dynamic Programming vs Greedy Method vs Divide & Conquer:

- In all methods the problem is divided into subproblem;
- **Greedy Method:** Building up of the solution to the problem is done in step-wise manner (incrementally) by applying local options only (local optimality).
- **Divide & conquer:** Breaking up a problem into separate problems (independent), then solve each subproblem separately (i.e. independently) & combine the solution of subproblems to get the solution of original problem.
- **Dynamic Programming:** Breaking up of a problem into a series of overlapping subproblems & building up solution of larger & larger subproblems.



Topic : Dynamic Programming: (DP)

- Unlike Divide and Conquer, D.P typically involves solving all subproblems, rather than a small portion of subproblem.
- D.P tends to solve each subproblem only once since the results of the subproblems are stored, which are used later again when required. This is going to reduce the computation drastically. (In most case, the complexity)

Ex- Fib { Brute force: $O(2^n)$
D.P Implementation: $O(n)$



Topic : Dynamic Programming: (DP)

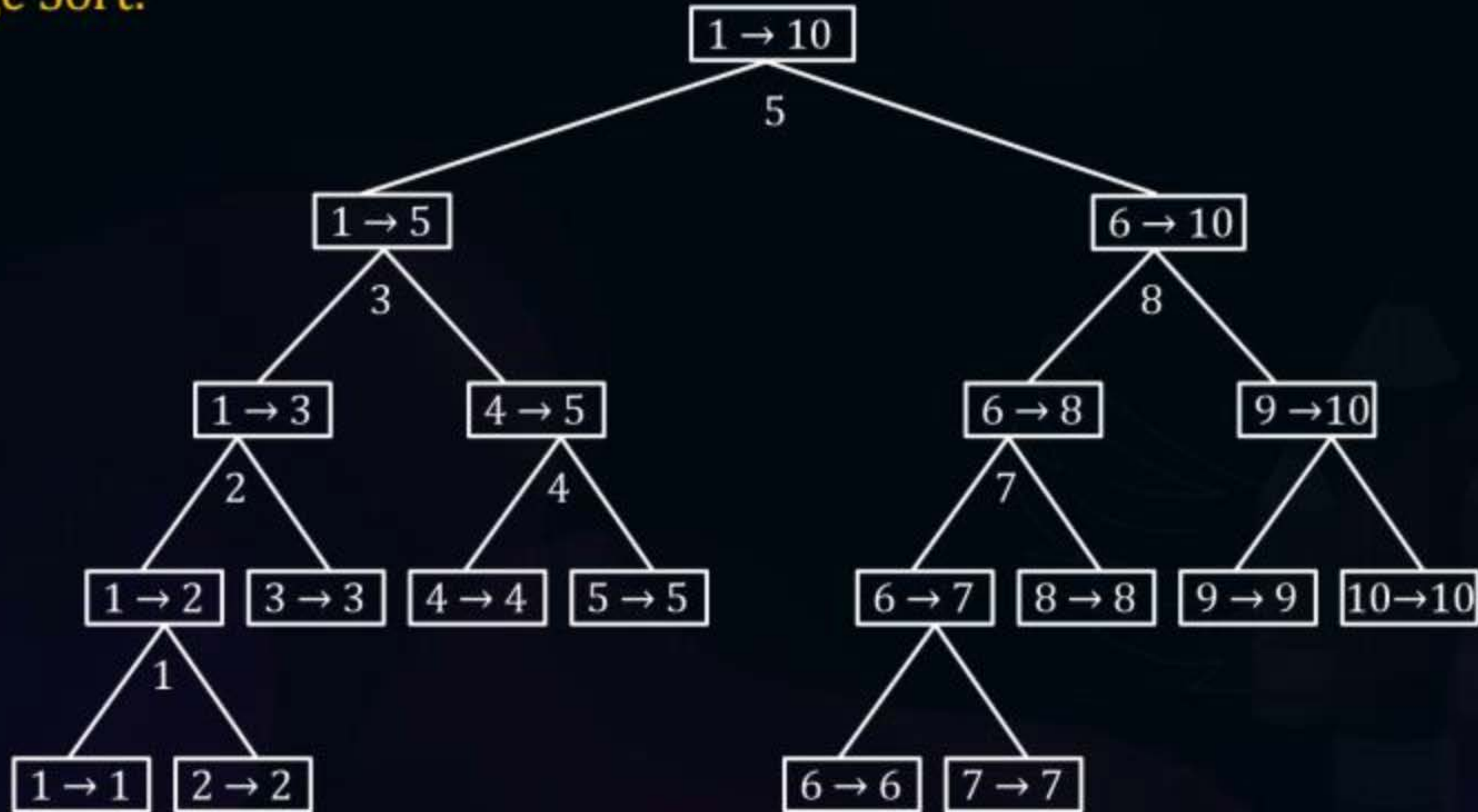
Imp. Diff. D_nC and DP:

1. DnC: independent sub-problems some separately
2. DP: Overlapping Sub- problems



Topic : Dynamic Programming: (DP)

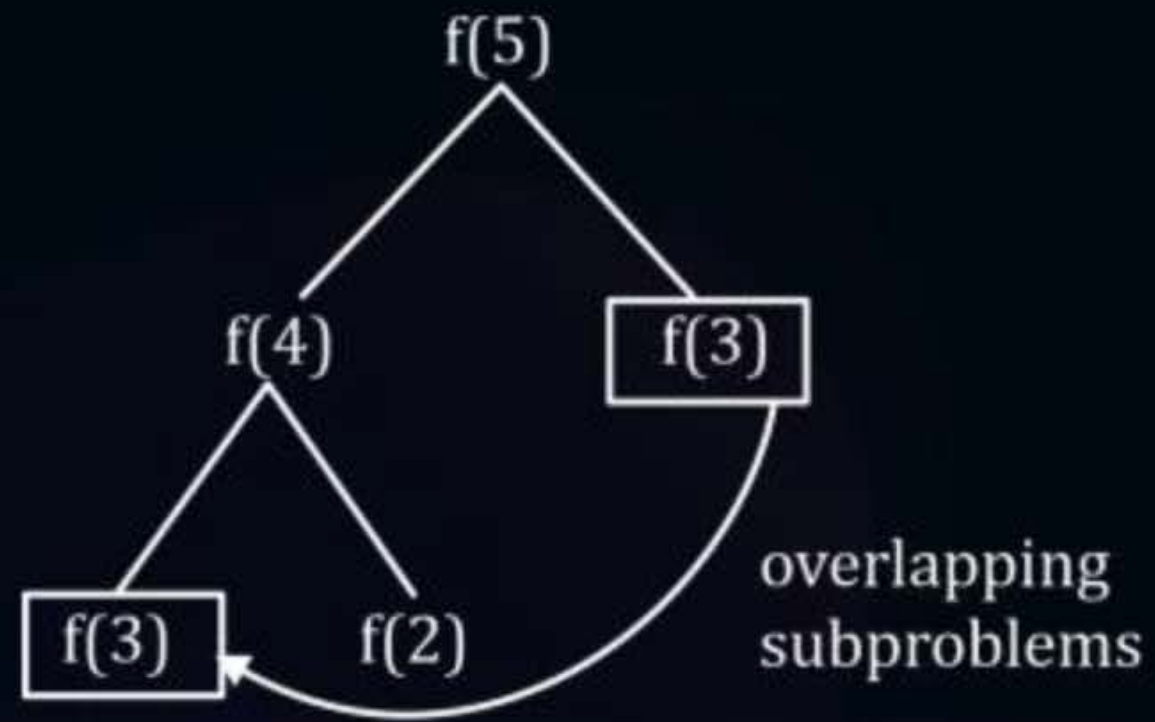
Merge Sort:





Topic : Dynamic Programming: (DP)

DP $\rightarrow$ Fibonacci



Application of DP:



Topic : Dynamic Programming: (DP)

Single source Shortest path (SSSP)

1. Dijkstra → Greedy ✓
2. Bellman ford → DP

Distinction between both SSSP algorithms and which to use when?



Topic : Dynamic Programming: (DP)

1. Dijkstra's SSSP (greedy algo) always gives optimal solution to the SSSP problems, provided all the edges in the given graph are of +ve weight.
2. If the graph has any -ve weight edge, then Dijkstra SSSP algo may or May not give optimal solution. (Path costs)



Topic : Dynamic Programming: (DP)

3. If the graph has 0 or more -ve weight edge But NOT a -ve weight cycle, then the bellman ford SSSP algo (DP based) always gives optimal solution to SSSP problem.
4. If the graph has any -ve weight cycle that is reachable from source, then No algorithm works.



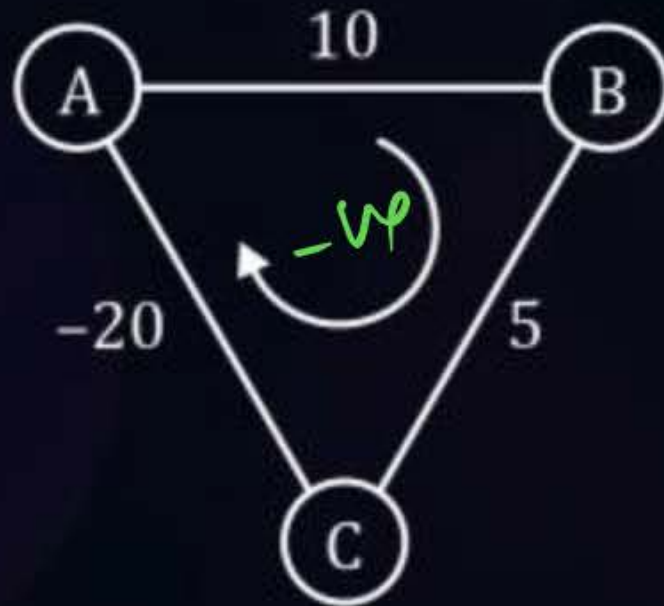
Topic : Dynamic Programming: (DP)

-ve weight cycle?

$$\begin{aligned}\text{Net weight of cycle} &= 10 + 5 - 20 \\ &= -5\end{aligned}$$

(-ve weight cycle)

* Min ^{cost}~~with~~ path is not defined.



$A \rightarrow B?$

10?

$A \rightarrow B: 10$

$A \rightarrow B \rightarrow C \rightarrow A \rightarrow B$

$-5 + 10 = 5$

$10 \rightarrow 5 \rightarrow 0 \rightarrow -5 \rightarrow -10$

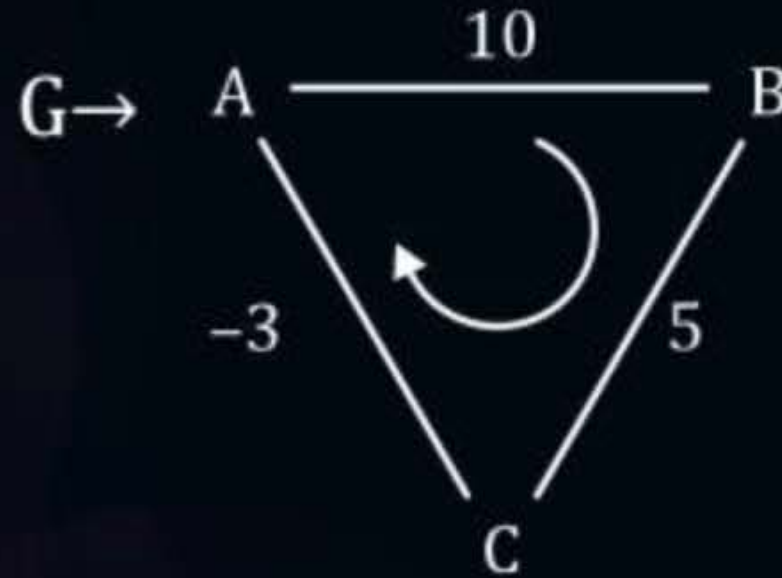


Topic : Dynamic Programming: (DP)

Eg.1. Graph with -ve weight edge but ^{-ve wt} No ~~required~~ ~~with~~ cycle.

$$\text{Net weight} = 10 + 5 - 3$$

$$= 12$$



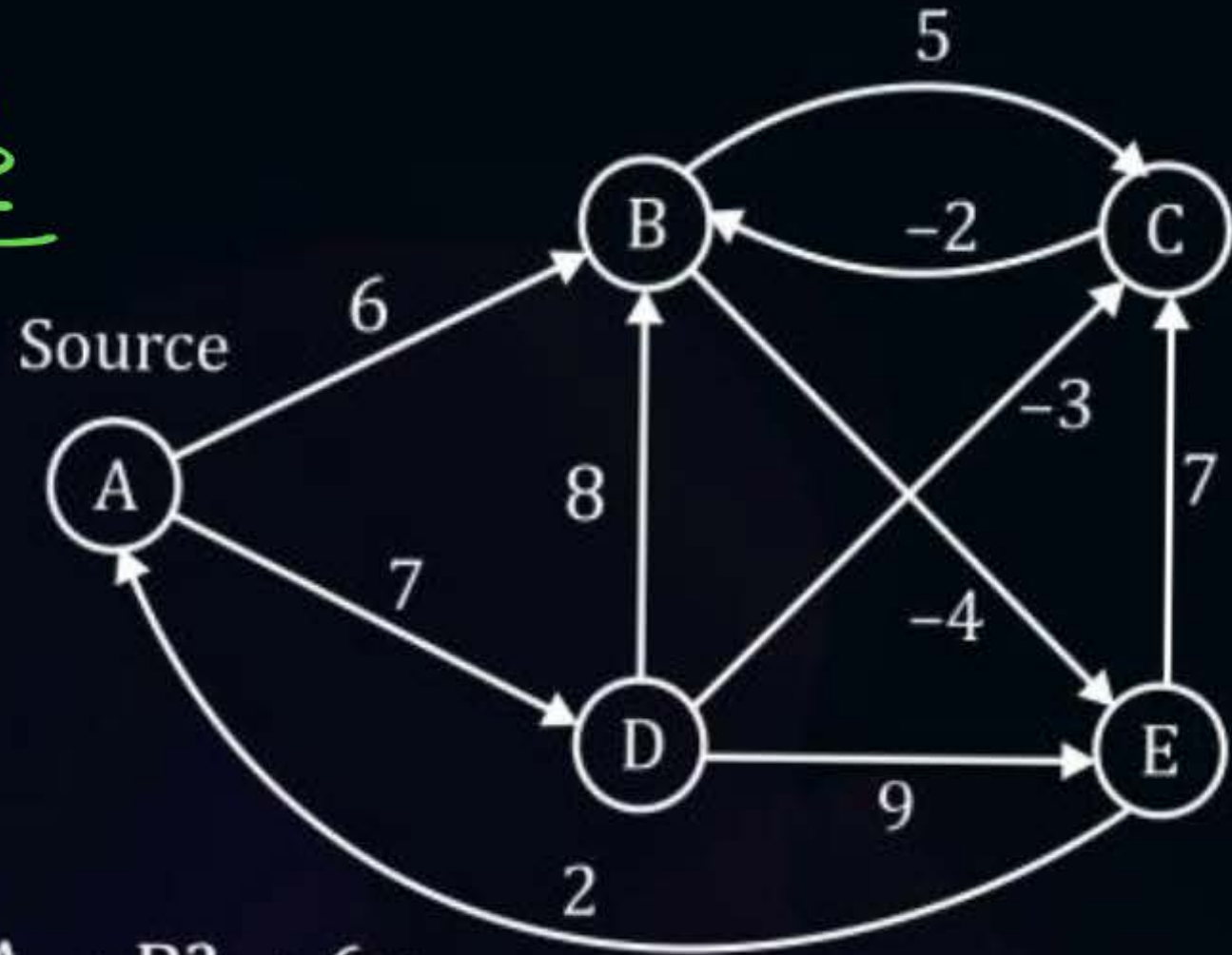
Dijkstra SSSP → Not guarantee (-ve weight edge)



Topic : Dynamic Programming: (DP)

Imp Given: G

A → B

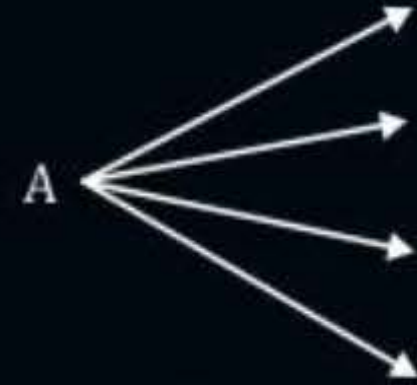


A → B? → 6 ×

A → D → C → B ⇒ $7 + (-3) + (-2)$

$= 7 - 5 = 2$ {Actual Minimum}

Apply Dijkstra SSSP Algo



	A	B	C	D	E
A →	0	6	4	7	2
		×	✓	✓	×

A → E : 2X

A → D → C → B → E ⇒ $7 + (-3) + (-2) + (-4)$
 $= 7 - 9 = -2$ {Actual min}



Topic : Dynamic Programming: (DP)

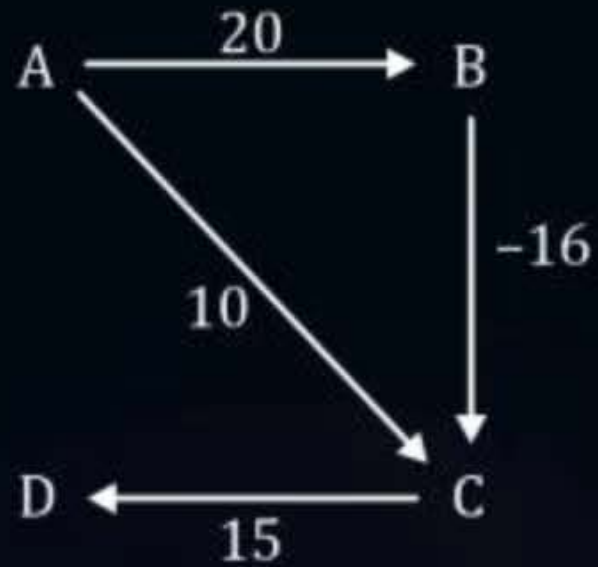
IMP Note:

1. In previous e.g. Dijkstra SSSP failed to give optimal path cost to few vertices, because in Dijkstra, once the vertex is selected is not further considered to be relaxed.
2. In Bellman ford, the relaxation is carried out w.r.t. to the edges in multiple iterations $(n-1)$ iterations.

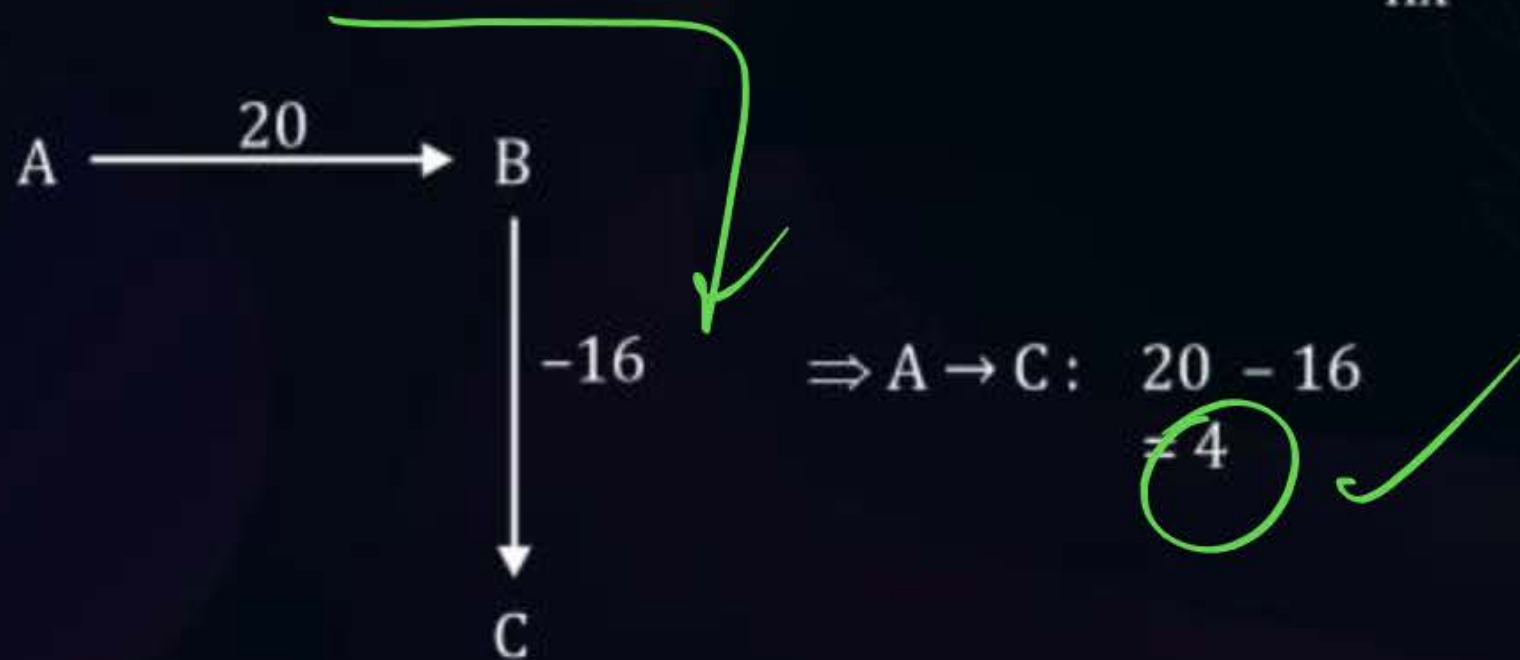
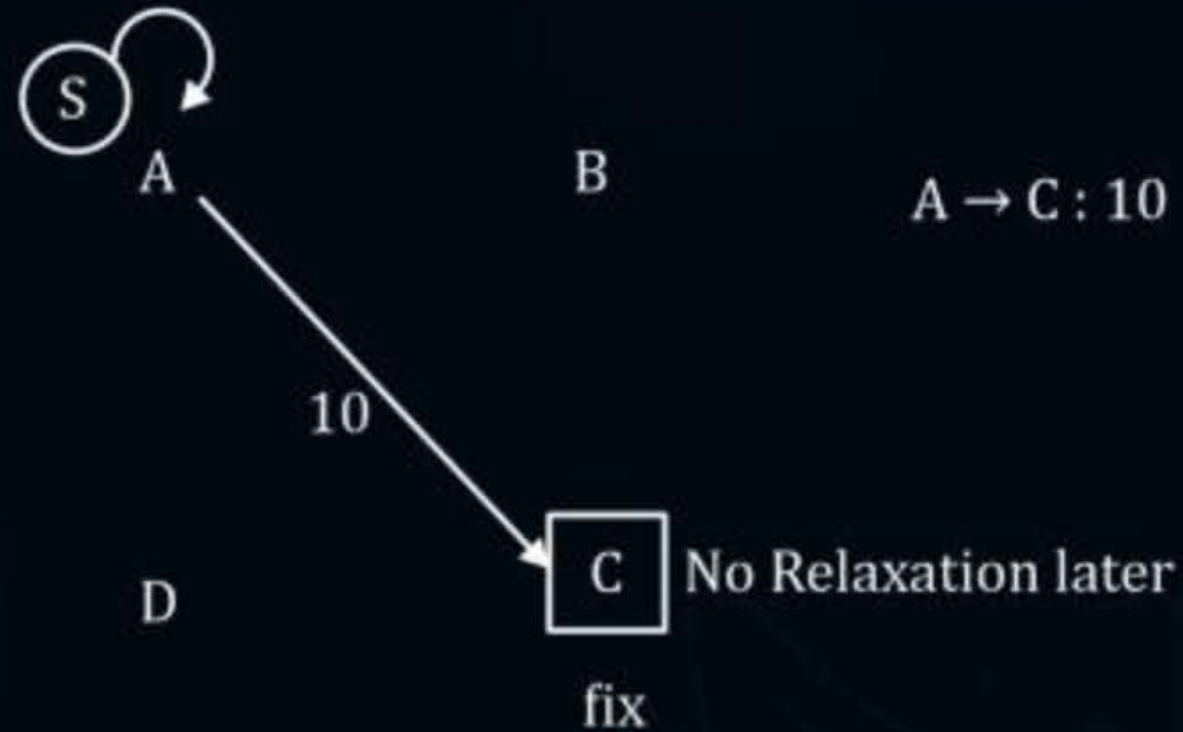


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Idea:-

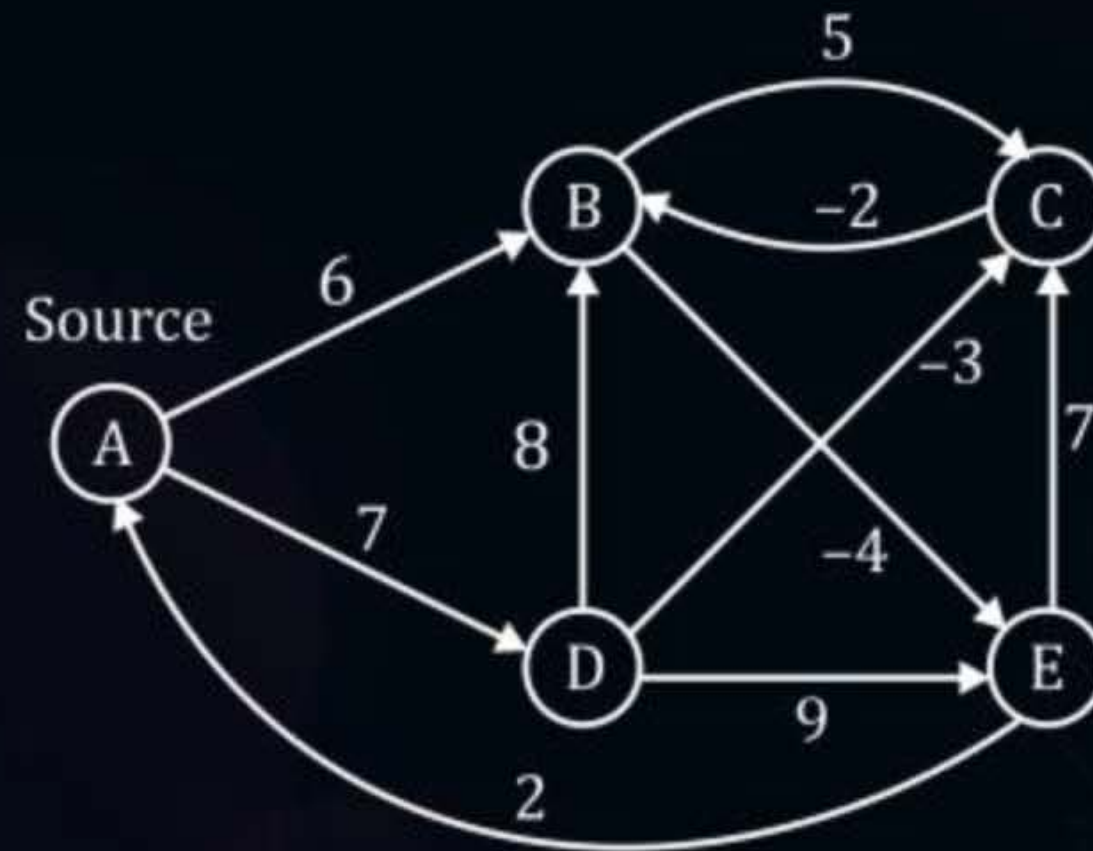


Dijkstra SSSP





Topic : Dynamic Programming: (DP)



$n = 5$

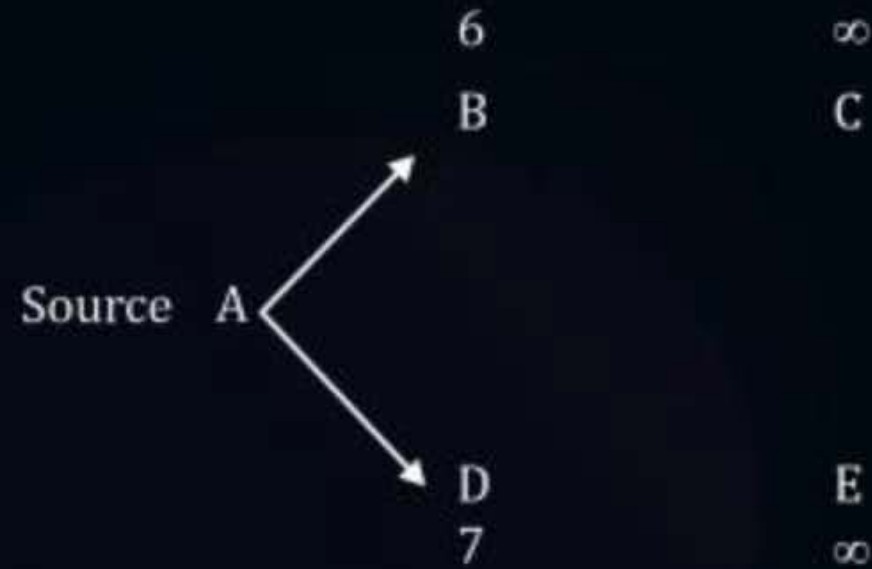
Iteration: $n - 1 = 4$



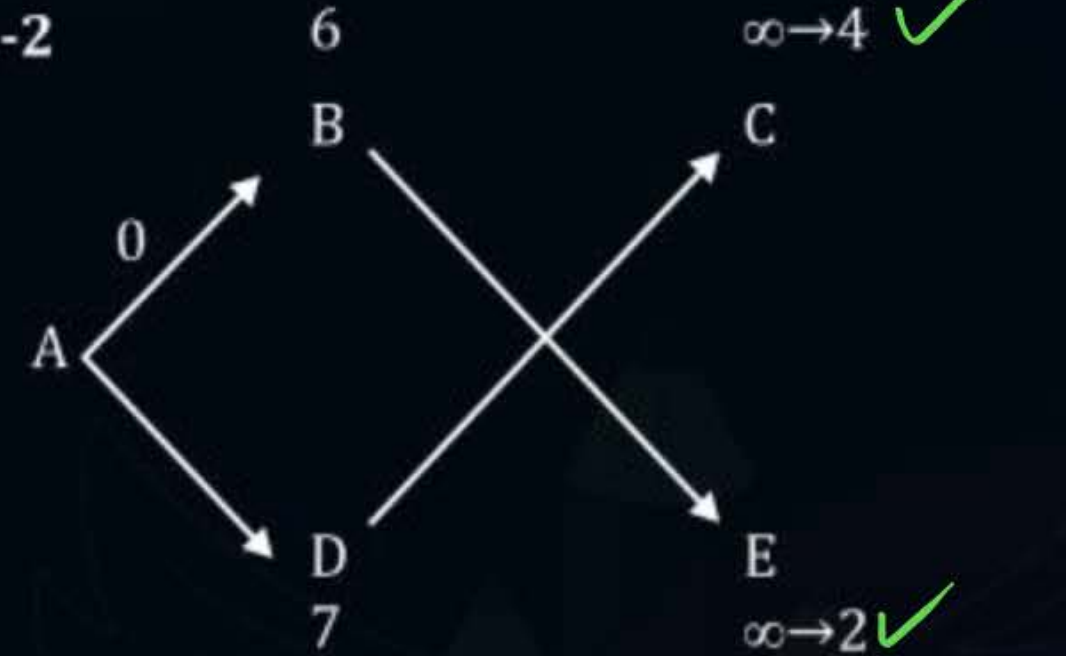
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Bellman ford Walk through

Iter.-1



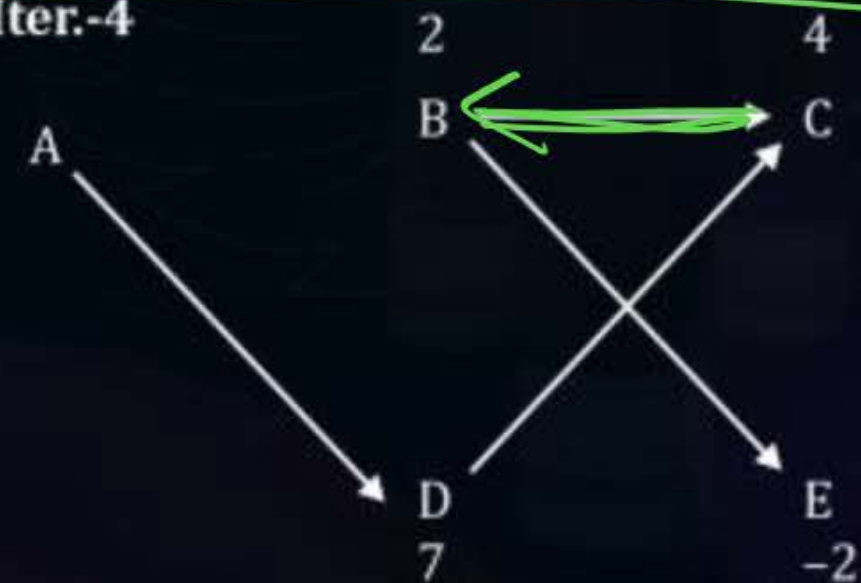
Iter.-2



Iter.-3



Iter.-4





Topic : Dynamic Programming: (DP)

Dj →

A → B : 6

A → C : 4

A → D : 7

A → E : 2

Bf

A → B : 2

A → C : 4

A → D : 7

A → E : -2

Imp. Observation:

Dijkstra Greedy SSSP algorithm failed to give us the shortest path to few vertices. But bellman ford SSSP given shortest paths (optimal solution) to all the vertices. (Even through -ve weight edge are present but no -ve weight cycle.)



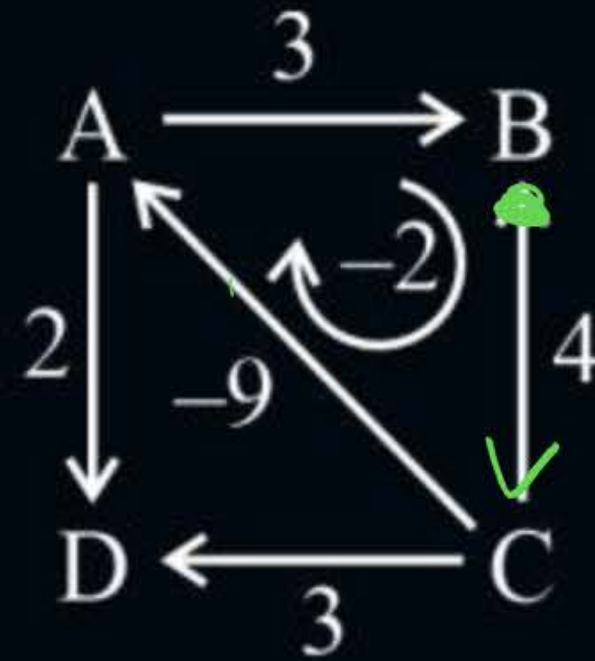
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Bellman Ford:

$$n = 4$$

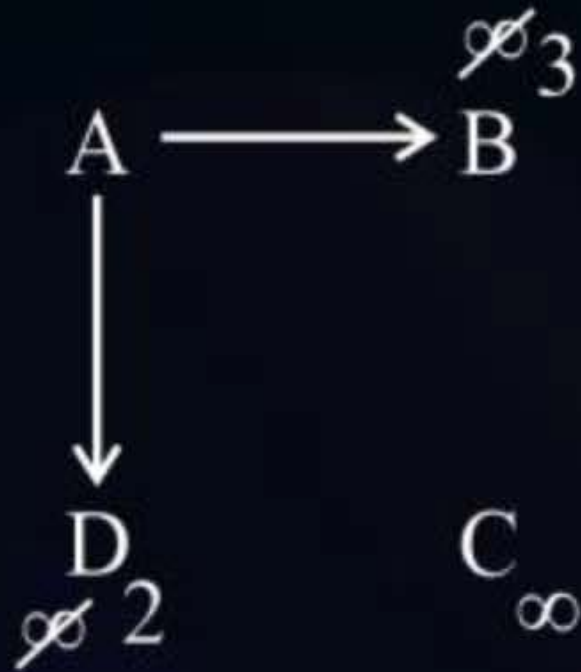
$$n - 1 = 3$$



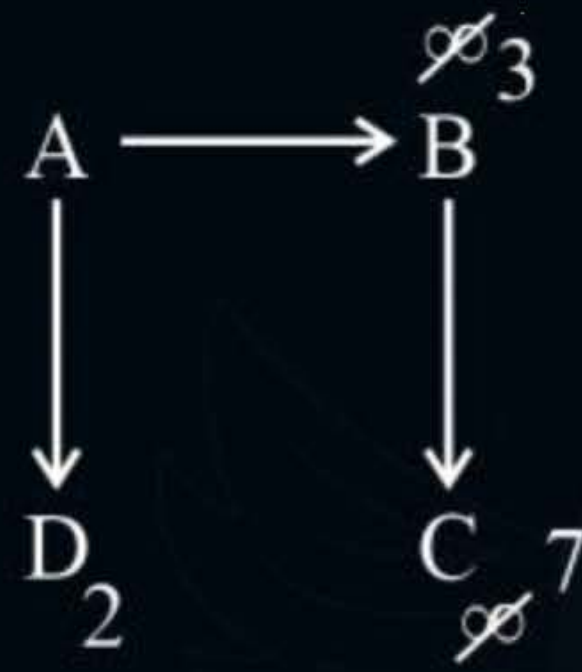


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Iteration 1:



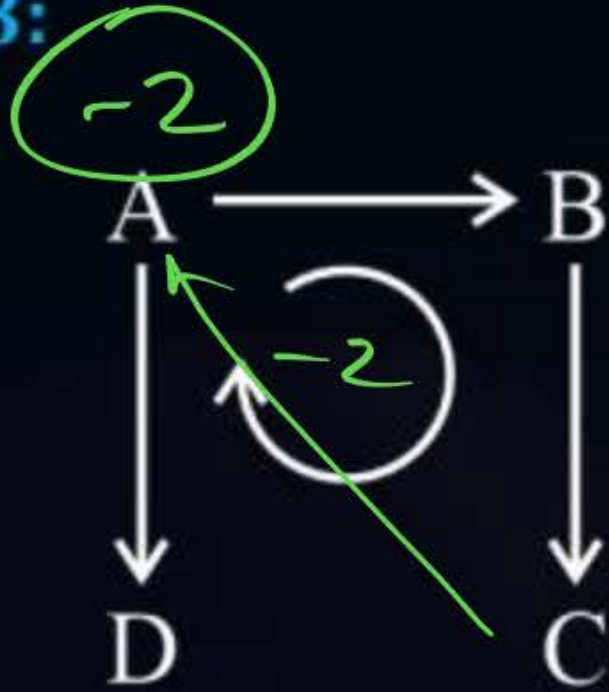
Iteration 2:





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Iteration 3:



In above example, as there is a negative ^{wt} ~~at~~ cycle reachable from source, so even Bellman Ford fails



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Algorithm Bellman-Fort (G, w, s)

1. Initialize-Single-Source(G, s)
2. for $i \leftarrow 1$ to $|V[G]| - 1$
3. do for each edge $(u, v) \in E[G]$
4. do Relax(u, v, w)
5. for each edge $(u, v) \in E[G]$
6. do if $d[v] > d[u] + w(u, v)$
7. then return FALSE
8. return TRUE



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Initialize – Single- Source (G, s)

1. For each vertex $v \in V[G]$
2. do $d[v] \leftarrow \infty$
3. $\pi[v] \leftarrow \text{NIL}$
4. $d[s] \leftarrow 0$

Relax (u, , w)

1. if $d[v] > d[u] + w(u, v)$
2. then $d[v] \leftarrow d[u] + w(u, v)$
3. $\pi[v] \leftarrow u$



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Overall Time Complexity:

- Bellman Ford SSSP DP approach

$$TC \rightarrow O(n + n * e + e) = O(n * e)$$

#Q. The time complexity of Bellman Ford algo for a complete graph having n vertices is?

$$T_c: O(n * e)$$

$$e \approx O(n^2)$$

$$\therefore T_c: O(n * n^2) = \underline{\underline{O(n^3)}}$$



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Floyd Warshall Algorithm:

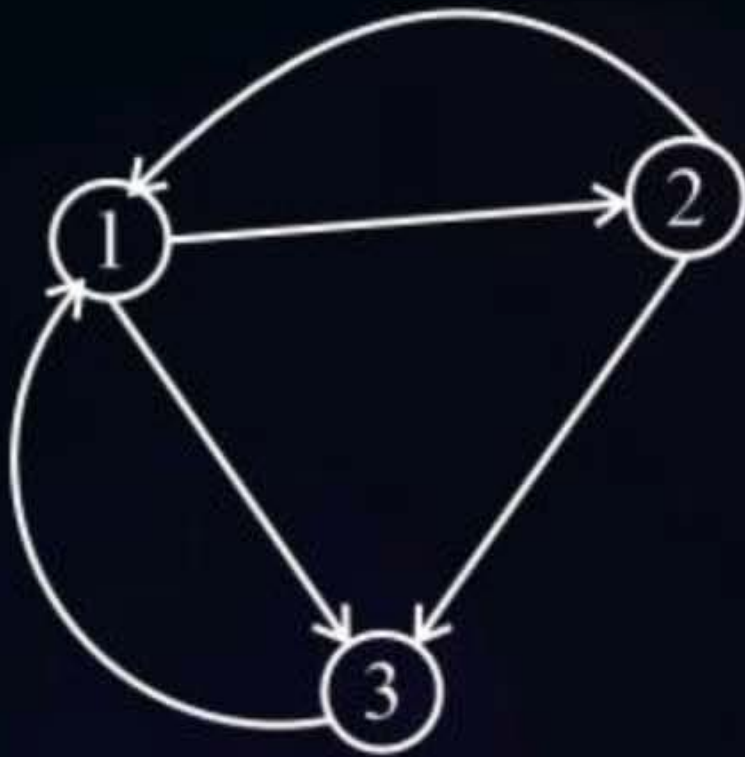
- For APSP (All Pairs Shortest Paths)



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Example:

Given, Cost adj matrix



C	1	2	3
1	0	4	11
2	6	0	2
3	3	∞	0





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DP based idea:

- Let $A^k(i, j)$ represent the cost of the path from a vertex 'i' (source) to some vertex 'j' (destination) with k being the highest intermediate vertex along the path from $i \rightarrow j$.



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$$A^k(i, j) = \min\{A^{k-1}(i, k) + A^{k-1}(k, j), A^{k-1}(i, j)\}$$

$$1 \leq k \leq n$$

$$A(i, j) = C(i, j)$$



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Algo FloydWarshall (G, C, n, e)

{

$A[1...n, 1...n] \longrightarrow SC: O(n^2)$

for (i=1; i<=n; i++)

for (j=1; j<=n; j++)

$A[i, j] = C[i, j]$

$\longrightarrow$ initialization

$\longrightarrow$ for (k=1; k<=n; k++)

for (i=1; i<=n; i++)

for (j=1; j<=n; j++) ✓

$A[i, j] = \min(A[i, j], A[i, k] + A[k, j])$

$\longrightarrow$ up dation

}

$a = a + 1$



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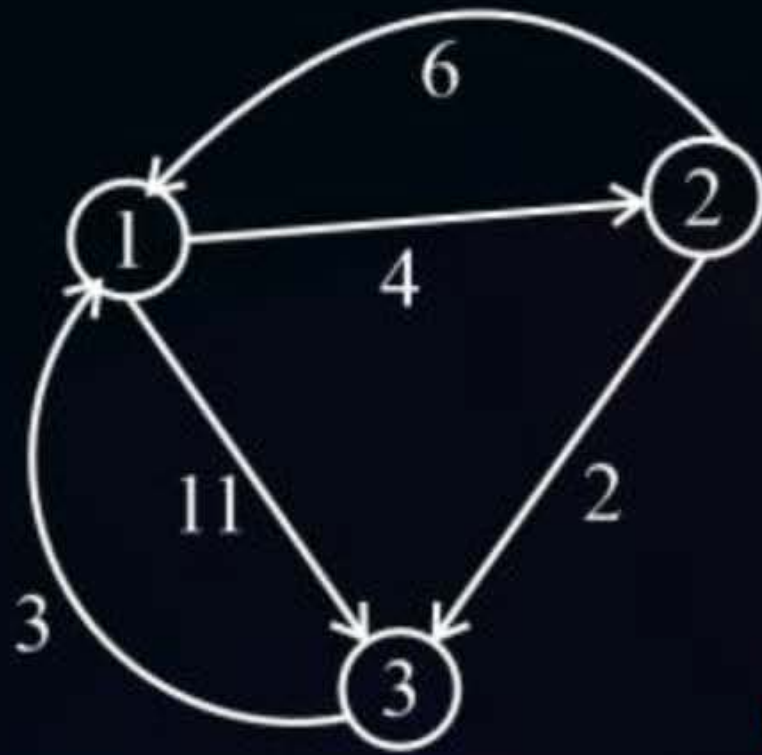
Time and Space Complexity Analysis:

- 1) Time Complexity = $O(n^3)$
- 2) Space Complexity = $O(n^2)$



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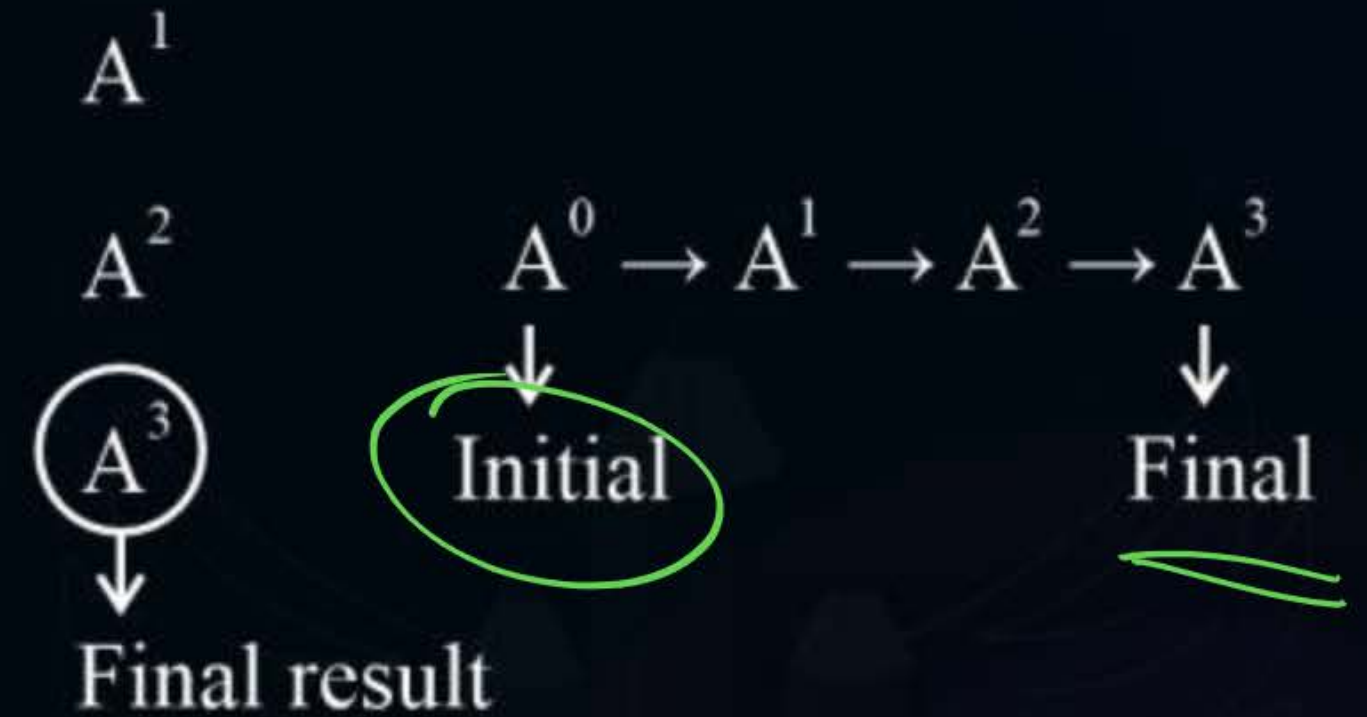
Code Walkthrough:



Bottom-up

C	1	2	3
1	0	4	11
2	6	0	2
3	3	∞	0

Tabulation





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Initialization, $A^0 = C$

A^0	1	2	3
1	0	4	11
2	6	0	2
3	3	∞	0

Intermediate, $k = 1$

A^1	1	2	3
1	0	4	11
2	6	0	2
3	3	7	0

Explanation:

$$3 \rightarrow 2 : A^0 : \infty$$

$$3 \rightarrow 1 \rightarrow 2$$

$$A^0[3, 1] + A^0[1, 2] = 3 + 4 = 7$$

1 → 3

1-2-3
 $4+2=6$



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$k = 2$

A^2	1	2	3
1	0	4	6
2	6	0	2
3	3	7	0

Explanation:

$1 \rightarrow 3 : A^1 : 11$

$1 \rightarrow 2 \rightarrow 3$

$4 + 2 = 6$

$k = 3$

Final APSP Soln

A^3	1	2	3
1	0	4	6
2	5	0	2
3	3	7	0

Explanation:

$2 \rightarrow 1 : 6$

$2 \rightarrow 3 \rightarrow 1$

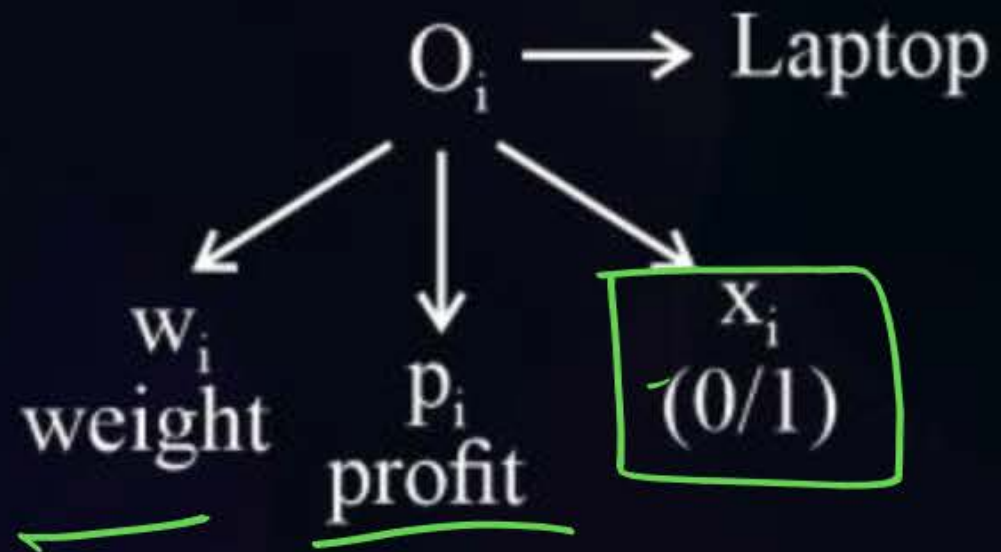
$A^2[2, 3] + A^2[3, 1] = 2 + 3 = 5$



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0/1 Knapsack (Binary Knapsack):

- Knapsack Capacity : M
- no. of objects → n





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Objective Function:

$$\max \sum_{i=1}^n \underline{p_i * x_i}$$

Such that:

$$\boxed{\sum w_i * x_i \leq M}$$

$$\underline{x_i = \{0, 1\}}$$



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Dynamic Programming Approach:

- Let 01 Knap(n, M) represent profit with n-objects and Knapsack of Capacity M.





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Recurrence:

HW

01 Knap(n, M) = 01 Knap($n-1, M$), $w_n > M$ → Exclude O_n (no option)

$$\underline{01 \text{ Knap}(n, M)} = \underline{\max} \begin{cases} 01 \text{ Knap}(n-1, M) \\ 01 \text{ Knap}(n-1, M - w_n) + p_n \end{cases}$$

$w_n \leq M$

Initialization,

01 Knap(n, M) = 0, if $n = 0$ or $M = 0$



2 mins Summary



Topic

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BF

FW

0/1 Knapsack



THANK - YOU